Pre-Calculus 11 - Cumulative Exam Part 2
Version 1 - With Calculator

Instructions, read carefully.

1. The formula sheet may be detached but must be handed in with the exam.
2. There are 9 total pages, including this front page and the formula page.
3. Final answers should be exact values or accurate to **two decimal places**.
4. Clearly show your steps or justify your reasoning in the space provided for full marks.
5. Read the box below and then write your name, student number, and signature.

St. George's School - Academic Integrity

Academic Dishonesty includes any conduct with the intent to gain an unfair advantage in connection with an academic assessment. Academic Dishonesty can occur in many ways. Common forms of Academic Dishonesty include, but are not limited to, cheating, falsification, plagiarism, and tampering. Cheating occurs when an individual undermines the integrity of an assessment.

I recognize that the writing of this examination requires an exemplary standard of honesty.

Name : _____ **Student Number :** _____

Signature : _____

Teacher Use only

Total: / 30

Quadratic Functions

1. [1 point] What is the vertex of the quadratic function $y = 3(x-4)^2 + 7$?

   A. $(-4, 7)$

   B. $(4, 7)$

   C. $(4, -7)$

   D. $(-4, -7)$

2. [1 point] Which of the following quadratic functions opens downwards and has a y-intercept of 2?

   A. $y = 2x^2 - x - 1$

   B. $y = x^2 + 3x + 2$

   C. $y = -(x-2)^2$

   D. $y = -x^2 + 3x + 2$

3. [1 point] A parabola has x-intercepts at $x = -2$ and $x = 8$. What is the equation of its axis of symmetry?

   A. $x = 4$

   B. $x = 5$

   C. $x = 6$

   D. $x = 3$

4. [1 point] Which quadratic function has a maximum value of 5?

   A. $f(x) = -5x^2 + 1$

   B. $g(x) = (x-1)^2 + 5$

   C. $h(x) = -(x-1)^2 + 5$

   D. $j(x) = -(x-5)^2 + 1$

5. [2 points] Convert the quadratic function $y = 2x^2 + 16x + 27$ from standard form to **vertex form**. Show your steps.

6. [2 points] Determine the equation of a quadratic function in vertex form that has its vertex at $(-3, 4)$ and passes through the point $(-1, -8)$.

7. [2 points] Determine the **exact** x-intercepts of the quadratic function $y = -3(x-2)^2 + 15$.

8. [2 points] A farmer has 120 meters of fencing to build a rectangular enclosure against an existing stone wall. Only three sides of the rectangle need fencing. Determine the dimensions that will **maximize** the enclosed area.

9. [2 points] A company sells custom mugs. If they charge \$10 per mug, they sell 200 mugs a month. Market research indicates that for every \$1 increase in price, they will sell 10 fewer mugs. Determine the price that maximizes their monthly revenue.

10. [1 point] State the domain and range of the quadratic function $y = -\frac{1}{2}(x+4)^2 + 9$.

Quadratic Equations

11. [1 point] What does a discriminant that is strictly **less than zero** indicate about a quadratic equation?

    A. It has exactly one real solution.

    B. It has two distinct rational solutions.

    C. It has two distinct irrational solutions.

    D. It has no real solutions.

12. [1 point] What are the solutions to $(2x-1)(3x+4)=0$?

    A. $x=-\frac{1}{2},\frac{4}{3}$

    B. $x=\frac{1}{2},-\frac{4}{3}$

    C. $x=-\frac{1}{2},-\frac{4}{3}$

    D. $x=\frac{1}{2},\frac{4}{3}$

13. [1 point] What are the solutions to the quadratic equation $2x^2-18=0$?

    A. $x=9$

    B. $x=3$

    C. $x=3, x=-3$

    D. $x=9, x=-9$

14. [1 point] Which solving method is the most efficient and direct way to solve the equation $(x-5)^2=12$?

    A. Factoring

    B. The Square Root Principle

    C. Completing the Square

    D. The Quadratic Formula

15. [2 points] Solve the following quadratic equation algebraically by **factoring**.

$$3x^2 - 10x + 8 = 0$$

16. [2 points] Solve the following quadratic equation algebraically.

$$2(x + 4)^2 - 50 = 0$$

17. [2 points] Solve the following quadratic equation. Leave your answer in exact radical form.

$$x^2 + 6x - 2 = 0$$

18. [2 points] A toy rocket is launched straight upward from a platform. Its height, h, in meters above the ground after t seconds is modelled by the quadratic function $h(t) = -4.9t^2 + 20t + 15$. Determine how long it takes for the rocket to hit the ground. Round your final answer to the nearest hundredth of a second.

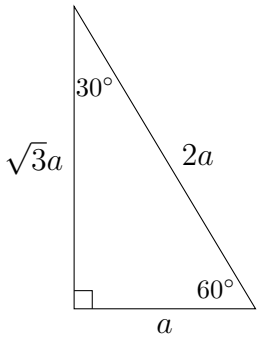
19. [1 point] Calculate the **discriminant** of the equation $3x^2 - 8x + 5 = 0$ and state the number of real roots it has.

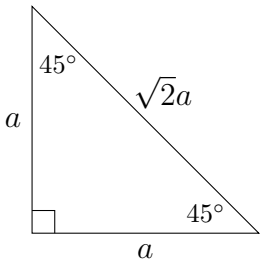
20. [2 points] The length of a rectangular garden is 4 meters longer than its width. The total area of the garden is exactly 45 square meters. Write a quadratic equation to model this situation, and solve for the exact width of the garden.

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Formula Page

Trigonometry





$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A \qquad \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Quadratic Equations

$$a^2 - b^2 = (a + b)(a - b)$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Quadratic Functions

$$y = a(x - h)^2 + k$$

$$y = ax^2 + bx + c$$

You may detach this page. Please return it with your exam.